

Surround speaker placement

- **5.1 System:** Place surrounds 2 ft above ear level, 90°–110° to the listening position.
- **7.1 System:** Add additional rear speakers directly behind the listening area.

4. Subwoofer placement and calibration

Proper subwoofer placement transforms boomy, uneven bass into tight, room-filling low frequencies. Unlike higher-frequency drivers, subwoofers are largely omnidirectional – the ear cannot localize bass as easily – so you have flexibility in placement.

1. Front corner start

- **Why Corner?** Corners reinforce bass by coupling boundary reflections, often yielding higher output for the same amplifier power.
- **How to Place:** Position the subwoofer a few inches away from the wall in one of the front corners, avoiding direct contact to reduce boundary-induced peaks.

2. The subwoofer crawl

- **Purpose:** Identify your room’s “sweet spot” for bass uniformity.
- **Steps:**
  - Temporarily place the subwoofer at your main listening position at normal ear height.
  - Play a bass-heavy test tone or music track with deep, varying frequencies.
  - Crawl on your hands and knees around the room perimeter, listening for where the bass sounds most balanced and flat.
  - Mark that spot and relocate the subwoofer there.

3. Multiple subwoofers for even coverage

- **Why Multiple?** Using two or more subwoofers in different corners or mid-wall positions excites room modes more evenly and smooths frequency response due to modal overlap.
- **Connection Methods:**
  - Daisy-chain via LFE line out.

- Split a receiver’s dual subwoofer outputs.
- Use wireless subwoofer adapters for flexible placement.

4. Phase and crossover settings

- **Crossover Frequency:** Set the subwoofer’s low-pass filter to match your main speakers’ low-frequency limit (typically 80 Hz) to ensure a seamless transition.
- **Phase Alignment:** Adjust the phase knob (0°–180°) while listening to pink noise or test tones until the bass blends without cancellations at your seating position.

5. Room EQ and calibration

- **Automated Calibration:** Many AV receivers and soundbars include room EQ (e.g., Audyssey, Dirac Live), which measures in-room response via a microphone and applies corrective equalization.
- **Manual EQ:** Use a parametric equalizer to tame specific room peaks and dips, focusing on ±3 dB adjustments at problematic frequencies.

6. Verification with measurement tools

- **SPL Meter Apps:** Validate level matching by measuring subwoofer output at the listening position, targeting ±2 dB relative to other channels.
- **Room Correction Software:** Tools like REW (Room EQ Wizard) can generate frequency response graphs, guiding placement tweaks and EQ adjustments.

5. Integrating components and fine-tuning

With soundbar or speaker placement and subwoofer calibration complete, it’s time to integrate all elements and fine-tune your system to the room.

System integration checklist

- **Speaker Polarity:** Confirm all speakers and subwoofers are wired with correct polarity to avoid phase cancellations.
- **Input Prioritization:** Route game consoles, Blu-ray players,

| REAL-WORLD AUDIO SETUP EXAMPLES                                                                                                                                      |                                        |                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------|
| Room Type                                                                                                                                                            | Configuration                          | Notes                                                                                   |
| Apartment Living                                                                                                                                                     | 2.1 Soundbar + Wireless Subwoofer      | Place bar under TV, subwoofer in front corner; use eARC; enable soundbar’s night mode.  |
| Dedicated Home Theater                                                                                                                                               | 5.1.2 Dolby Atmos Speakers + Dual Subs | Speakers in equilateral layout; subs in opposite front corners; Dirac Live calibration. |
| Music-Focused Setup                                                                                                                                                  | Stereo Bookshelf Speakers + Subwoofer  | Bookshelf speakers on stands, 2 ft from walls; subwoofer via subwoofer crawl placement. |
| Each example underscores matching gear to room constraints and content habits. Calibration and thoughtful placement ensure you reap the benefits of your investment. |                                        |                                                                                         |